No of Pages : 3 Course Code : 23MX24

Roll No:
(To be filled in by the candidate)

PSG COLLEGE OF TECHNOLOGY, COIMBATORE 641 004
SEMESTER EXAMINATIONS, JUNE 2025
MCA Semester: 2
23MX24 ENTERPRISE COMPUTING USING FULL STACK

Time: 3 Hours **Maximum Marks: 100**

INSTRUCTIONS:

1. Answer **ALL** questions. Each question carries 25 Marks.
2. Subdivision (a) carries 3 marks, subdivision (b) carries 10 marks and subdivision (c) carries 12 marks each.
3. Course Outcome: Table

Qn.1	CO1	Qn.2	CO2	Qn.3	CO3	Qn.4	CO4

COs	At the end of learning this course, students will be able to
CO1	Exemplify the concepts of Enterprise, Enterprise architecture and applications using Java and Java Script Technology Patterns and Frameworks
CO2	Develop enterprise Web Database Application using SQL, NOSQL and JAVA Frameworks
CO3	Develop Enterprise Application using Java script based Frameworks
CO4	Develop inter-enterprise applications using SOAP and REST based Web services and Micro Services

		Question	(BTL)
1.	a)	How do patterns and frameworks enhance or optimize the development of enterprise applications?	(L4)
	b)	Scenario: A university wants to automate the admission process including application submission, merit list generation, and seat allocation across departments. As an architect, suggest a full-stack architecture (e.g., Java and/or JavaScript-based) that maintains modularity and allows departmental autonomy. Explain your choice of layers and technology stack with justifications.	(L5)
	c)	Formulate the requirements and challenges to design the following enterprise application that would integrate the disparate applications within a hospital into a coherent whole. Also, construct a Full Stack architecture that supports Separation of Concerns in this enterprise application while realizing these requirements and challenges. Over many years, a hospital has developed a number of separate systems for patient administration. There is a system for handling patient registration, one for appointment scheduling, and another for medical records, a billing system, a pharmacy management system, and separate systems for laboratory and radiology departments. Some departments still use paper-based systems for certain processes. As a result, patient data is fragmented, leading to inefficiencies and potential errors in patient care. The hospital seeks an enterprise computing system that would integrate these disparate applications into a coherent whole. This should be a multi-tier architecture which is flexible in its support of various client programs handling interaction with users.	(L5)

2. a) What is a prepared statement in JDBC, and how does it mitigate SQL injection risks? (L2)

b)i) In the context of Java JDBC, compare and contrast the Execute, ExecuteUpdate, and ExecuteQuery methods in terms of their functionality, return types, and use cases. (5) (L4)

ii) Explain the need for NO SQL in Enterprise Applications and also compare the role SQL and NO SQL in Data enabling aspects of Enterprise Applications. (L4)

c) i) 1) Discuss JDBC application architecture and explain how JDBC increases the longevity of Java enterprise DB applications code over embedded form like SQLJ. (6) (L3)

2) Discuss the advanced JDBC features 'Connection Pooling' and 'Distributed Transaction,' emphasizing their implementation in the JDBC context and their practical applications in business use cases such as high-traffic e-commerce platforms. (6) (L4)

(OR)

ii) 1) Write a Java Program that to connect with database using the following information (3) (L3)

Driver : com.mysql.jdbc.Driver

DBURL: jdbc:mysql:/ /localhost:3306/test

User Name: 23MXIans

Password: 123@24

2) Consider the following Table Student (L3)

Columns: Roll No: Char(8), Name varchar (25), Course varchar (25) , Faculty varchar(30) and Marks: int

Write JDBC statements to retrieve all faculty members whose courses with above 70% students with marks are greater than 60%. (3)

3) Evaluate the server-side rendering mechanism of JSP (Java Server Pages) processing by analyzing its key components and their interactions. Synthesize how these components collectively contribute to optimizing dynamic web content generation, and justify the advantages of JSP's approach over alternative server-side technologies in terms of performance and scalability. (6) (L4)

3. a) Justify JavaScript's suitability for use across the key layers of a full stack enterprise application, highlighting its benefits for developing scalable and efficient enterprise solutions (L4)

b) Through a case study of an e-commerce web platform, compare the advantages and trade-offs of Single-Page Applications (SPAs) versus Multi-Page Applications (MPAs) in web development, focusing on their impact on user experience and development efficiency. (L5)

c) Consider the following scenario: (L5)

A start-up is developing an e-learning platform where users can enroll in courses, access video lectures, submit assignments, and participate in discussion forums. The platform must support real-time notifications for assignment deadlines and forum replies, and ensure security against unauthorized access.

- Propose a JavaScript based Full Stack architecture (MERN or MEAN) for this platform.
- Discuss the implementation of React components for the front-end, Node.js/Express for the back-end, and MongoDB for data storage.
- Explain how the architecture supports real-time features and security measures like CAPTCHA for user registration.

4. a) Compare and contrast APIs, Web Services, and Micro services in the context of inter-enterprise applications, focusing on their respective advantages and limitations for enabling seamless cross-organizational system integration. (L4)

b) Analyze the use of Restful web services versus SOAP-based services for an enterprise application integrating a logistics company's tracking system with third-party delivery partners. Discuss the role of JSON, HTTP methods, and routing in Restful services. (L5)

c) i) 1) Given a scenario where an enterprise application needs to handle millions of concurrent user interactions, analyze how REST's architectural constraints help or hinder the system's scalability. (6) (L5)

2) Describe how you would design, develop, deploy, and consume Java Restful web services to enable secure and efficient data exchange for a scenario where a healthcare system needs to integrate with external patient record databases. (6) (L5)

(OR)

ii) 1) Consider a scenario where an insurance management system needs to integrate with external policy verification services, illustrate how to design, develop, deploy, and consume Java SOAP web services to ensure secure and reliable data exchange. (6) (L5)

2) Assess how the SOAP web service standard stack facilitates interoperability and compatibility across systems running on different platforms and programming languages in a multinational retail inventory management application. (6) (L5)

CO : COURSE OUTCOME; BTL : BLOOM'S TAXONOMY LEVEL

/END/

FD/RL